Week 9 Assignment

Everything is a linear model

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 10's lecture (Monday). See the syllabus for the late policy.

Load the packages you need first:

```
library(tidyverse)
library(marginaleffects)
```

Part A: Group comparison. Using the built-in `ToothGrowth` data, compare tooth length (`len`) between the two supplements (`supp`) at the highest dose (`dose == 2`).

1. Run a two-sample t-test (`var.equal = TRUE`).
2. Run the same comparison with `lm()`, and show that the p-value matches.
3. Use `avg_comparisons()` and `avg_predictions()` on your linear model to report the group means and their difference. Write one sentence reporting the result with its 95% confidence interval.

Part B: Continuous predictor. Using the built-in `trees` dataset:

4. Fit `lm(Volume ~ Height)`, and use `avg_slopes()` to report the slope.
5. Show that the standardized slope (use `scale()` on both variables) equals `cor(trees$Height, trees$Volume)`.
6. Check the diagnostics for the model in question 4. Does a straight line look reasonable?

Part C: Concept.

7. In 2–3 sentences, explain why a paired t-test is equivalent to an intercept-only linear model, and what the data in that model are.
8. `trees` is observational. What would need to be true about how data were collected before you could claim "increasing height *causes* increased volume"?